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Santanu Sarkar

Frontend Architect · Senior Frontend Engineer

PROFESSIONAL SUMMARY

Frontend Architect with 14+ years of experience delivering enterprise web platforms for clients in India and Singapore. Specialises in React, Next.js (App Router) and Vue 3 component systems on headless Drupal, with a focus on performance, accessibility and maintainable architecture. Outside client work, designed, built and operates three personal multi-tenant products end to end, from data model and authentication to testing and deployment.

TECHNICAL SKILLS

Languages & Markup	HTML5, CSS3 / SCSS, JavaScript (ES6+), TypeScript
Frameworks	React, Next.js (App Router), Vue 3, Nuxt
Styling & Design	Tailwind CSS, Bootstrap 5, GSAP, Figma
Backend & Platform	Headless Drupal, REST / JSON:API, SEO / Core Web Vitals, AI dev tools
Full-stack (personal)	Server Actions / route handlers, Prisma, Drizzle ORM, Neon Postgres, Auth.js, Vitest
AI & Tooling	Claude Code, GitHub Copilot, Cursor, Git / GitHub / Bitbucket, Chrome DevTools

PROFESSIONAL EXPERIENCE

Frontend Architect
Mar 2015 – Present

Webspiders Pvt. Ltd · Kolkata & Singapore · joined as Frontend Developer

- Led end-to-end frontend for enterprise and SaaS products.
- Translated Figma systems into modular React and Vue components.
- Built headless Drupal + React / Next.js frontends on REST and JSON:API.
- Raised Lighthouse scores and tightened Core Web Vitals across projects.
- Mentored junior developers on component architecture and code quality.
- Implemented SEO foundations: semantic HTML, structured metadata, Open Graph and server-side rendering.
- Worked with UX designers, backend engineers and product managers through UAT and production releases.

HTML Developer
Sep 2014 – Feb 2015

Budli.in

- Built mobile-optimized landing pages for paid campaigns.
- Improved product page UX, navigation and checkout flow.
- Fixed frontend and cross-browser issues across the site.

HTML Developer
Apr 2012 – Sep 2014

Delgence

- Built client websites with a focus on responsive layout.
- Applied accessibility and performance best practices.

PERSONAL PRODUCTS

Three products built solo under Nexvora Technologies, my own MSME in Kolkata (2026), using AI coding assistants (Claude Code). Owned the product design, architecture and data model, directed the implementation, reviewed and tested the output, ran QA, and handled deployment and production incidents.

Halkhata – QR ordering and a credit ledger for local shops
Live · first real shop onboarded

halkhata.nexvoratechnologies.co.in | Next.js 15, TypeScript, Prisma, Neon Postgres, PWA, Web Push

- Server-authoritative money in integer paise, idempotent orders under a unique key, and race-safe stock decrements inside the sale transaction.
- Stateless HMAC-signed sessions verified at the Edge and re-verified against the database before any write; credential versioning revokes every device on a PIN reset.
- Wrote up a production data-loss incident and added a guard that refuses schema commands against production.

EduFlow – Fees, attendance and WhatsApp reminders for coaching centres — Live · real centres

eduflow.nexvoratechnologies.co.in | Next.js 15, Drizzle ORM, Neon Postgres, Auth.js, Server Actions, PGlite tests

- Four-role multi-tenant SaaS with tenant isolation enforced on every read and write and data scoped by role on the server.
- Diagnosed and fixed a failing billing cron (Postgres 42P10 on a partial-index ON CONFLICT) that had stopped monthly fee generation for live centres.
- Integration tests run on in-memory Postgres (PGlite) built from the real SQL migrations and gate every deploy.

HealthLocker – A multi-tenant family health record platform — Pilot · demo data only

health-locker-chi.vercel.app | Next.js 16, Prisma, Neon Postgres, Column encryption, GitHub Actions CI, Capacitor

- Server-first Next.js monolith on 49 Prisma models; tenant derived from the session, AES-GCM column encryption and a nonce-based CSP.
- Found and closed two cross-tenant authorization holes and proved the fix with a role-by-guard test matrix.
- CI on every push: lint, typecheck, 250 unit tests, about 100 integration blocks, a production build and a smoke render of every page per role.

SELECTED CLIENT PROJECTS

IPI Singapore ipi-singapore.org — 2024 / 25

Innovation marketplace portal — frontend architected end-to-end in Next.js App Router on headless Drupal. React · Next.js · Drupal · Tailwind · GSAP

SpiderX.ai spiderx.ai — 2023 / 24

Marketing site for a generative voice-agents platform — custom interactions, pixel-perfect delivery. HTML5 · Bootstrap 5 · jQuery · Drupal

PeerlessOne.in peerlessone.in — 2023

Financial services platform on a Vue 3 / Nuxt frontend integrated with Drupal CMS. Vue 3 · Nuxt · Bootstrap 5 · Drupal

SGInnovate sginnovate.com

Responsive frontend and custom interactions for the SGInnovate platform. HTML5 · Bootstrap 5 · jQuery · Drupal

Affinidi affinidi.com

Reusable React components integrated with a Drupal CMS backend. React · Drupal

EDUCATION & ADDITIONAL

Higher Secondary (Class XII) – Ultadanga Govt. High School, Kolkata — 2001

Secondary (Class X) – Ultadanga Govt. High School, Kolkata — 1999

Certifications: Web Design, Digital Marketing

Languages: Bengali (native), Hindi (fluent), English (professional working)